



A systematic review of resting-state EEG across the menstrual cycle and its mental health relevance

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Abstract

Purpose This systematic review aims to synthesize current evidence on resting-state EEG activity across the menstrual cycle. Specifically, it explores whether cyclical hormonal fluctuations of estradiol and progesterone are associated with electrophysiological changes and evaluates the potential of resting-state EEG as a possible biomarker for neurophysiological changes related to mood, cognition, and emotional well-being across the healthy menstrual cycle.

Methods A systematic search of PubMed, Web of Science, Scopus, SpringerLink, and PsycINFO was conducted to January 2025, identifying 23 relevant studies. PRISMA guidelines were followed to extract relevant articles, and we assessed the quality of the evidence using the GRADE approach.

Results A total of 23 studies met the inclusion criteria. The most convergent findings show that the alpha EEG activity in frontal, parietal, and temporal brain areas tends to decrease, and the theta activity tends to increase during the late follicular phase (characterized by high estradiol and low progesterone) compared to the luteal phase (when estradiol level is lower and progesterone is higher). This resting-state electrophysiological activity could reflect greater attentional efficiency, emotional well-being, and a reduction in self-referential processing in the days surrounding ovulation. Findings for delta, beta, and gamma bands remain inconclusive.

Conclusions This review suggests that resting-state EEG may serve as a potential biomarker of hormone-related changes in brain activity, offering preliminary insights for further research and early detection of hormone-sensitive mood disorders.

Highlights

- Resting-state EEG activity changes across the menstrual cycle and may be hormonally influenced.
- Alpha EEG activity tends to decrease and theta activity tends to increase during the late follicular phase, compared with the luteal phase, suggesting improved attention and emotional regulation in the days surrounding ovulation.
- These findings tentatively point to a possible hormone-related neural basis for mood disturbances, reinforcing the importance of cycle phase in women's mental health research.

Keywords Resting-state EEG · Menstrual cycle · Women's mental health · Progesterone · Estradiol

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Background

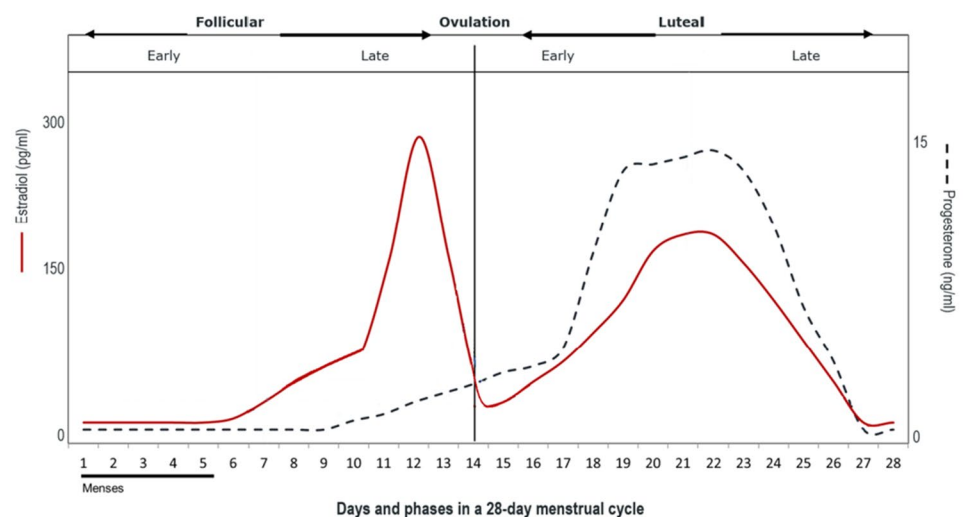
According to United Nations estimates, there were approximately 1.9 billion women of reproductive age (15–49 years) worldwide in 2021. Of these, an estimated 1.1 billion are naturally cycling. This represents approximately 58% of the global population of women of reproductive age, while the remaining 42% use either modern or traditional methods of contraception (United Nations, 2022). The menstrual cycle is a 28–32 day neuroendocrine process marked by fluctuating levels of estradiol and progesterone, which influence various physiological and psychological functions (Dubol et al. 2021). The menstrual (M) phase begins with low estradiol and progesterone. In the early follicular (F) phase, progesterone remains low while estradiol rises, peaking in the late F phase before the ovulatory (O) phase. The early luteal (L) phase shows rising progesterone and a smaller estradiol peak in the mid L phase. Both hormones decline in the late L phase and reach their lowest in the premenstrual phase (Roney and Simmons 2013) (see Fig. 1).

Beyond their reproductive roles, these hormones exert significant neuromodulatory effects on the central nervous system, influencing synaptic plasticity, neurotransmission, and large-scale neural networks involved in emotion, cognition, and behavior (Hidalgo-Lopez et al. 2021; Pritschet et al. 2020; Barth et al. 2015). As such, the menstrual cycle is increasingly recognized not only as a biological rhythm but also as a critical factor modulating brain function and mental health (Liparoti et al. 2021). Recently, the menstrual cycle has even been proposed as an overlooked “vital sign” in psychiatry, reinforcing its relevance for women’s mental health research (Marwick et al. 2025). Emerging evidence suggests that the late F phase is associated with enhanced cognitive and affective functioning, including improved verbal memory, attentional control, and emotional well-being.

Conversely, the L phase has been linked to increased affective sensitivity, irritability, and, in susceptible individuals, the emergence of clinically significant symptoms as observed in premenstrual dysphoric disorder (PMDD) (Comasco and Sundström-Poromaa 2015). At the moment, several neuroimaging studies have summarized the impact of sex hormones on the brains of women with healthy menstrual cycle (e.g., Dubol et al. 2021; Rehbein et al. 2021; Toffoletto et al. 2014); however, a comprehensive review of electrophysiological findings in healthy women throughout the menstrual cycle is still lacking. Therefore, understanding the electrophysiological correlates of estradiol and progesterone hormonal shifts is essential for elucidating the biological basis of hormone-sensitive mental health symptoms in women.

Electroencephalographic (EEG) recording taken in a resting state is a powerful method to assess *in vivo* spontaneous electrical activity of the brain, providing information about its functioning and processing (Biasucci et al. 2019). In addition to low cost and being non-invasive, EEG offers a temporal resolution in milliseconds of voltage fluctuations of the electrical activity of the cerebral cortex, being detected by multiple electrodes placed on the scalp. EEG activity typically is grouped into delta, theta, alpha, beta, and gamma oscillations. According to Başar and colleagues (2013), spontaneous oscillatory activity of the EEG regulates the majority of brain functions, and the measurement of the quantitative EEG parameters (e.g., amplitude, frequency, relative and absolute power) serves as a control criterion of brain responsiveness. Resting-state EEG activity has the potential to function as an electrophysiological biomarker in the clinic to provide, for example, an electrophysiological-hormonal diagnosis in a population of women with PMDD (Kaltsouni et al. 2024). It is relevant to consider that some electrophysiological studies that evaluate the

Fig. 1 Schematic displaying the hormonal fluctuations of estradiol and progesterone across a 28-day menstrual cycle, with ovulation occurring on day 14



effect of estradiol and progesterone hormones on resting-state EEG activity diverge in terms of methodology applied. For example, the type of hormonal assessment, the days that define the phases of menstrual cycle, the menstrual cycle phase used for comparisons, the cortical area for the EEG recordings, and the EEG analysis. Besides, some research could present some bias when they include women without distinguishing between the menstrual cycle phases or when they include solely men and exclude women due to the fluctuation of sex hormones across the menstrual cycle. Thus, the current systematic review emphasizes that the menstrual cycle phases are a key variable that should be considered in order to ensure a more accurate interpretation of cortical activity both in clinical practice and in neuroscience research. Therefore, the present systematic review provides a comprehensive and integrative knowledge of resting-state EEG activity in reproductive-aged women associated with fluctuations of estradiol and progesterone throughout the healthy menstrual cycle.

Methods

Search strategy and information sources

We followed the Preferred Reported Items for Systematic Reviews and Meta-Analyses (PRISMA) 2021 guidelines (Page et al. 2021). A complete search of the literature published in English until January 2025 was conducted in PubMed, SpringerLink, SciELO, Google academic, and Scopus. The following search terms and their combinations were used: ('menstrual cycle', OR 'menstrual phase', OR 'follicular phase', OR 'luteal phase' OR 'ovulatory phase', OR 'female sex hormones', OR 'estradiol', OR 'estrogen', OR 'progesterone') AND ('EEG', OR 'electrical activity of the brain', OR 'electroencephalogram', OR 'resting-state EEG', OR 'EEG at rest condition').

Eligibility criteria

Following title and abstract screening, we excluded papers upon full-text review if they failed to meet the following criteria: EEG study, EEG recordings at rest condition, healthy women in reproductive age (17–49 years, House et al. 2012) with regular menstrual cycle (28–32 days), determination of the menstrual phase by the counting method and/or confirmation of the menstrual cycle phase through the analysis of blood, salivary, or urinary hormone levels, and English language. We included studies even if the EEG resting state was not the primary endpoint but only a secondary objective measure in the investigations. In contrast, articles based on magnetoencephalography, evoked potential, magnetic

resonance imaging, diffusion tensor imaging, positron emission tomography, and EEG studies in women with psychological disorders (e.g., major depressive disorder) or neurological disorders (e.g., epilepsy) were excluded for the present review. In addition, EEG studies in non-human models (e.g., in primates or rodents) were also excluded. Finally, only spontaneous resting-state EEG data were considered, while dynamic EEG responses to various tasks or cues were not considered for the current review.

Selection and data collection process

Two independent reviewers (M.D.G.-G. and R.M.-C.) performed the research, reading first the title and the abstract, then the full text if relevant. After this first step, they grouped the articles meeting the inclusion criteria and reached a consensus on which to include. In the event of disagreement, a third reviewer was consulted (L.H.-L.) to decide whether or not to include the article in the review.

Collected items

We extracted the following information for each study when available: sample size, age, hormonal measurement, menstrual cycle phase, women's health status, EEG recording condition, EEG recording area, bands analyzed (delta, theta, alpha, beta, and gamma), and research findings. Table 1 shows the definitions of the variables considered to analyze the EEG activity. To facilitate comparison across studies, menstrual cycle phases reported in the literature were mapped onto four standardized categories: menstrual, follicular, ovulatory, and luteal (Table 2). This categorization was based on well-established hormonal and physiological characteristics of the menstrual cycle: the F phase encompasses the early and late F stages, the O phase corresponds to the LH surge and ovulation, and the L phase covers the post-ovulatory secretory stage (Roney and Simmons 2013). We retained the original terminology to see the precise labels used in each study, while the standardized phases enabled consistent synthesis of findings.

Quality assessment of included studies

Study quality was assessed by two reviewers (M.D.G.-G. and R.M.-C.), using a strategy based on the recommendations of the Grading of Recommendations Assessment Development and Evaluation (GRADE) working group (Guyatt et al. 2011). We used a modified version of the Downs and Black checklist to assess study quality (Downs and Black 1998) (Supplementary Material). The modified Downs and Black checklist comprised 15 outcomes from five domains: (1) reporting, (2) external validity, (3) internal validity—bias,

Table 1 Definition of the EEG variables extracted from included studies

Term	Unit	Definition
Frequency	Hz	It is the rate at which current changes direction per second. The frequency is measured in hertz (Hz), where 1 Hz is equal to 1 cycle per second.
Amplitude	μV	The maximum displacement of points on a wave, which you can think of as the degree or intensity of change.
Relative power	%	It is the percentage of the AP in one brain oscillation compared to the total power spectrum across all brain oscillations of interest.
Absolute power	μV^2 (or dB)	It is the total energy intensity of an electrode on a certain region at different frequency bands.
Dominant frequency or Peak frequency	Hz	Frequency at which the spectral component is the highest in a given frequency range.
Amplitude suppression magnitude (Berger Effect)	lg%	The natural logarithm of the percentage decrease in amplitude in response to eyes open.
Weighted frequency	Hz	It represents the product of multiplying the weights of each set of frequencies with their respective mean and dividing the result by the total sum of the frequency weights.
Individual alpha frequency (IAF)	Hz	The highest peak within a frequency window from 7 to 14 Hz.

Table 2 Standardized menstrual cycle phase labels derived from original study terminology

Standardized phase	Original terminology
Menstrual	Menstruation Menses
Follicular (early or late)	Follicular Proliferative Intermenstrual Preovulatory
Ovulatory	Ovulation Mid-cycle
Luteal (early or late)	Luteal Pre-menstrual Postovulation

(4) internal validity—confounding (selection bias), and (5) power. A maximum attainable score of 16 could be awarded, whereby study quality was categorized as follows: high (14–16), moderate (10–13), low (6–9), or very low (0–5). The results of the Downs and Black were used to assign an a priori quality rating to each study. Then, as previous studies did (Elliott-Sale et al. 2020; McNulty et al. 2020), a priori rating was either maintained or downgraded a level, based on the response to the following question: Q.1) Was the menstrual cycle phase confirmed using blood, saliva, or urine samples? If the authors reported using blood, saliva, or urine samples to confirm the menstrual cycle phase, the a priori

rating was maintained, and if not, the study was downgraded a level (e.g., a study that started out as high in quality but did not confirm the menstrual cycle phase using blood, saliva, or urine samples drops to moderate in quality). This distinction is important because hormonal verification provides an objective measure of cycle phase, ensuring more accurate classification of participants. In contrast, studies relying only on self-reported cycle dates may introduce misclassification bias, which can affect the reliability of findings. Therefore, studies were divided into those that included hormonal measurement and those that did not.

Analysis

Given the heterogeneity in the research (e.g., EEG recordings, definition of menstrual phases, sample size, etc.), the primary endpoint of our systematic study was the report of changes in EEG activity at rest in healthy menstrual cycle women of reproductive age. Likewise, we informed the association between resting-state EEG activity and sex hormones (estradiol and progesterone). To synthesize the evidence, we performed a study-by-study count of reported increases or decreases in EEG activity, considering each EEG band and menstrual phase according to the results described.

Results

Literature search

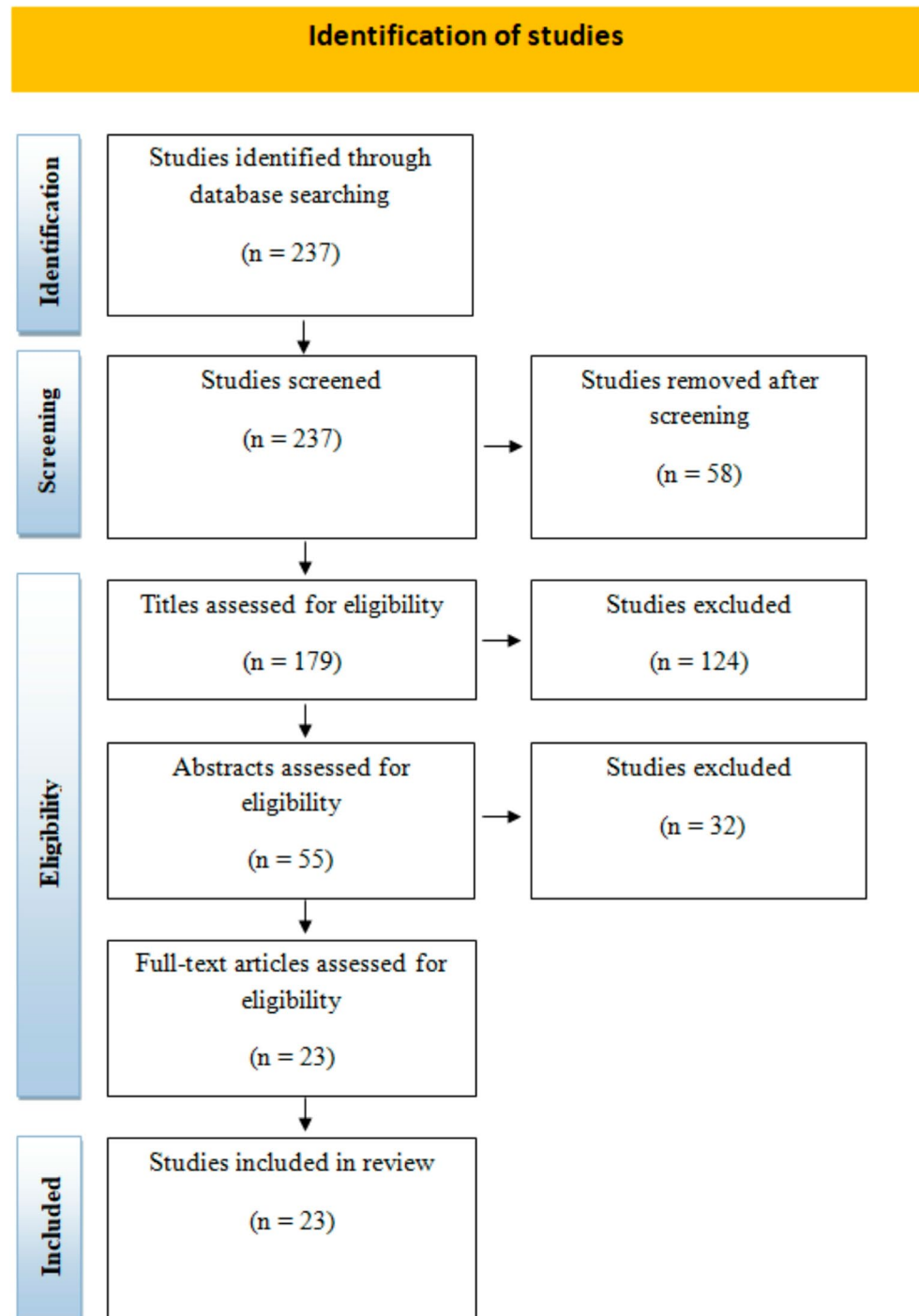
The literature search and selection of studies are presented in Fig. 2.

Study characteristics

Twenty-three studies with 573 women were included in this review (age range: 17–42; sample size range: 8–114). Of these, 10 studies (43.47%) did not include hormonal measurement (Table 3), whereas 13 studies (56.52%) did (Table 4).

Quality assessment of included studies

The quality classification of each study is presented in Table 5. Analysis of quality based on the entire evidence ($n=23$) was ascertained at the individual study level and according to the Downs and Black checklist, as well as the additional question regarding the menstrual cycle phase confirmation. The a priori quality from the 23 studies was classified as 47.8% high, 39.1% moderate, 13% low, and 0% very low. However, after the menstrual cycle phase verification by hormonal assay test,

Fig. 2 PRISMA flow diagram of study deletion

the final rating of quality was 39.1% high, 26% moderate, 26% low, and 8.7% very low (see Fig. 3).

Resting-state EEG activity in women with healthy menstrual cycles

The changes in the resting-state EEG activity across different phases of the menstrual cycle are presented in Table

3 for studies without hormonal measurement and in Table 4 for studies with hormonal measurement. Overall, studies with and without hormonal measurements show similar trends. Accordingly, Fig. 4 provides a general overview of EEG activity changes across the menstrual cycle, while Supplementary Figs. 5 and 6 present the detailed results for studies without and with hormonal measurements, respectively.

Table 3 Resting-state EEG activity through the healthy menstrual cycle without hormonal verification

Study	N	Age	Hormonal assessment	Original terminology of the menstrual cycle phase	Harmonized menstrual phase terminology	Status of women	EEG recordings condition	EEG recording areas	Band frequency analyzed	Results
Dusser de Barenne and Gibbs, 1942	11	Unspecified	No	Every day along of one menstrual cycle	M phase, F phase, O phase, L phase	Healthy women	At rest	Right frontal Parietal Occipital	Alpha (unspecified frequency range)	The onset of M phase was followed by slowing in alpha frequency.
Margerson et al. 1964	12	17–28	No	Intermenstrual (day 12 to 14 from the first day of bleeding) Premenstrual (one day before bleeding or the first day of bleeding)	Late F phase Late L phase	Healthy women with regular menstrual cycles	At rest with eyes closed	Occipital Temporal	Alpha (8–13 Hz)	No significant differences in the dominant frequency late F phase vs. late L phase.
Gautray, 1969	8	Unspecified	No	Proliferative (unspecified days) Ovulation (unspecified days)	F phase O phase	Healthy women with normal menstrual cycles	At rest with eyes closed	Frontal Temporal Occipital	Theta (4–8 Hz) Alpha (8–12 Hz)	Relative power of theta in the left frontotemporal during the F phase > O phase. Relative power of alpha in the posterior scalp in the F phase < O phase.
Vogel et al. 1971	14	18–21	No	Preovulation (8th day after first day bleeding) Postovulation (20th day after first day bleeding)	Late F phase L phase	Healthy women with regular menstrual cycles	At rest with eyes open with photic stimulation	Left and right occipital	Alpha (unspecified frequency range)	Alpha frequency in the late F phase < L phase.
Leary and Batho 1979	25	20–35	No	Premenstrual (one day before of menstruation) Mid-cycle (unspecified-day)	Late L phase O phase	Healthy women with regular menstrual cycles	At rest	Parietal Occipital	Alpha Theta (unspecified frequencies range)	Mean alpha frequency and mean alpha amplitude in the late L phase increase vs. O phase. Theta activity is twice as common in the late L phase vs. O phase.
Kaplan et al. 1990	20	18–30	No	Follicular (day 9 from the first day of bleeding) Luteal (day 23 from the first day of bleeding)	Late F phase L phase	Healthy women with regular menstrual cycle	At rest with eyes closed	Occipital	Alpha (7.5–11.5 Hz)	Peak frequency of alpha in the late F phase < L phase.

Table 3 (continued)

Study	N	Age	Hormonal assessment	Original terminology of the menstrual cycle phase	Harmonized menstrual phase terminology	Status of women	EEG recordings condition	EEG recording areas	Band frequency analyzed	Results
Solis-Ortiz et al., 1994	9	20–34	No	Ovulation (day 13–14 from the first day of bleeding) Postovulation (day 20–21 from the first day of bleeding) Premenstrual (day 24–25 from the first day of bleeding) Menstrual (day 1–2 day of bleeding)	O phase L phase Late L phase M phase	Healthy women with regular menstrual cycles	At rest with eyes closed and eyes open	Frontal Central Parietal Occipital	Delta (1.5–3.5 Hz) Theta (3.5–7.5 Hz) Alpha 1 (7.5–9.5 Hz) Alpha 2 (9.5–12.5 Hz) Beta 1 (12.5–17.5 Hz) Beta 2 (17.5–30 H)	Rest with eyes open: Absolute power of delta and alpha 1 at the left occipital area in the O phase < late L phase. Absolute power of beta 1 at right central area during the O phase < late L phase. Absolute power of alpha 1 at left frontal, beta 1 at right central and left parietal, and beta 2 at left central areas in the O phase < M phase. Delta absolute power in the occipital areas in the O phase < M phase. The absolute power of delta at right frontal and alpha 1 at frontal areas in the late L phase > L phase. The absolute power of theta at right frontal in the late L phase > M phase. Relative power of alpha 1 at right central and alpha 2 at right frontal in the L phase < late L phase. Rest with eyes closed: No significant difference in the absolute power in any oscillations. Relative power of alpha 2 at left frontal areas in the late L phase > L phase. Relative power of delta at right parietal areas in the M phase > O phase.

Table 3 (continued)

Study	<i>N</i>	Age	Hormonal assessment	Original terminology of the menstrual cycle phase	Harmonized menstrual phase terminology	Status of women	EEG recordings condition	EEG recording areas	Band frequency analyzed	Results
Kaneda et al. 1997	44	21	No	Follicular (day 1–13 from the first day of bleeding) Luteal (day 15–28 from the first day of bleeding)	Early and late F phases L phase	Healthy women with regular menstrual cycles	At rest with eyes closed	Occipital	Delta (2–3.75 Hz) Theta (4–7.75 Hz) Alpha 1 (8–9.75 Hz) Alpha 2 (10–12.75 Hz) Beta 1 (13–19.75 Hz) Beta 2 (20–30.00 Hz)	No significant difference in the absolute power in any oscillation between early and late F phases vs. L phase. A tendency for increased relative power of alpha and decreased relative power of beta in the LP vs. FP.
Vasil'eva, 2005	35	18–38	No	Ovulation (day 9–14 from the first day of bleeding) Menstruation (day 2–3 day of bleeding)	O phase M phase	Healthy women with regular menstrual cycles	At rest	Frontal Central Parietal Occipital	Alpha (8–13 Hz)	Alpha absolute power in central and mediotemporal areas during the O phase < M phase.
Mulik et al. 2017	20	17–22	No	Menstrual (the second or third day of the menstruation) Follicular (second day after menstruation) Preovulatory (4–6 days in respect to follicular phase) Ovulatory (on the day of the maximal LH concentration in urine at peak times) Luteal (in 5 days in respect to ovulatory phase) Premenstrual (6 days in respect to luteal phase)	M phase Early F phase Late F phase O phase L phase Late L phase	Healthy women with regular menstrual cycles	At rest with eyes closed	Occipital	Alpha (8–13 Hz)	A minimum value of IAF is observed in the M phase. IAF in L phase decreases vs. the beginning of the cycle.

M menstrual, *F* follicular, *O* ovulatory, *L* luteal, *IAF* individual alpha frequency

Table 4 Resting-state EEG activity through the healthy menstrual cycle with hormonal verification

Study	N	Age	Hormonal assessment	Menstrual cycle phase	Harmonized menstrual phase terminology	Status of women	EEG recordings condition	EEG recording areas	Band frequency analyzed	Results
Wuttke et al. 1975	16	18–25	E2 and P4	Menstruation (first day of bleeding) Ovulation (unspecified day) Luteal (unspecified day)	M phase O phase L phase	Women with regular menstrual cycles	At rest with eyes closed	Occipitocentral midline Left and right temporal	Alpha (8–13 Hz)	Alpha frequency at occipitocentral in the L phase > O phase > M phase.
Creutzfeldt et al. 1976	16	20–28	E2 and P4	Follicular (unspecified days) Luteal (unspecified days)	F phase L phase	Healthy women with spontaneous cycles	At rest with eyes closed	Occipitocentral Temporocentral	Theta (3.91–7.43 Hz) Alpha (7.82–14.08 Hz) Beta (14.47–25.02 Hz)	Theta power in the occipitocentral and temporocentral in the L phase < F phase. Mean theta frequency in the temporocentral in the L phase < F phase. Alpha power in temporocentral in the L phase < F phase. The weighted mean of the alpha band in the occipitocentral and temporocentral in the L phase > F phase. No significant differences in the beta power and the beta mean frequency in the occipitocentral and temporocentral in the L phase vs. F phase.
Becker et al. 1982	7	18–28	E2 and P4	Ovulation (day 0, determined by the LH peak) Follicular (days –8 to 0) Luteal (days 2 to 10) Menstrual (days 12, 14 and –10)	O phase Early and late F phases L phase M phase	Healthy women with spontaneous menstrual cycles	At rest with eyes closed	Occipitocentral Temporocentral	Theta (2.5–6 Hz) Alpha 0 (6.5–9 Hz) Alpha 1 (9.5–10.5 Hz) Alpha 2 (11–13 Hz) Beta 1 (13.5–22 Hz) Beta 2 (22.5–30 Hz)	Theta frequency at the occipitocentral area in the M phase > L phase. Negative correlation between theta frequency and P4 level. Mean theta power was lower during the F phase and higher during the L phase. Absolute power of alpha 0 in FP (i.e., at times of low P4) > L phase (i.e., at times with high P4). Absolute power of beta 1 and 2 during the early and late F phase > L phase. The power of beta 1 and 2 are higher at times of low P4.
Krug et al. 1999	16	21–33	E2 and P4	Ovulation (day 11–16 from the first day of bleeding) Midluteal (4 and 10 days prior to menses) Menstruation (day 2–5 from the first day of bleeding)	O phase L phase M phase	Healthy women with regular menstrual cycles	Mental relaxation with eyes open	Frontal Central Parietal	Delta (1.0–3.5 Hz) Theta (3.5–7.0 Hz) Alpha (8.0–12.1 Hz) Beta (14.1–30.1 Hz)	Alpha and theta power in frontal, central, and parietal areas during the L phase < O phase and M phase.

Table 4 (continued)

Study	N	Age	Hormonal assessment	Menstrual cycle phase	Harmonized menstrual phase terminology	Status of women	EEG recordings condition	EEG recording areas	Band frequency analyzed	Results
Solis-Ortiz et al. 2004	9	19–34	E2 and P4	Ovulation (day 13–14 from the first day of bleeding) Early luteal (day 20–21 from the first day of bleeding) Late luteal (day 24–25 from the first day of bleeding) Menstruation (day 1–2 day of bleeding)	O phase L phase Late L phase M phase	Healthy women with regular menstrual cycles	At rest with eyes open	Frontal Central	Delta (1.5–3.5 Hz) Theta (3.5–7.5 Hz) Alpha 1 (7.5–9.5 Hz) Alpha 2 (9.5–12.5 Hz) Beta 1 (12.5–17.5 Hz) Beta 2 (17.5–30 Hz)	Absolute power of alpha 1 and alpha 2 decreased whereas beta 1 increased during the L phase. Absolute power of alpha 1 and alpha 2 increased whereas beta 1 decreased during the late L phase and M phase. No significant difference in absolute power of delta, theta and beta 2 during the MC.
Solis-Ortiz et al., 2009	9	20–34	E2 and P4	Menstruation (day 1 and 3 of bleeding) Follicular (day 7 and 8 from the first day of bleeding) Ovulatory (day 13 and 14 from the first day of bleeding) Early luteal (day 20 and 21 from the first day of bleeding) Late luteal (day 24 and 25 from the first day of bleeding)	M phase Late F phase O phase L phase Late L phase	Healthy women with regular menstrual cycles	At rest with eyes open	Frontal Central Parietal Occipital	Delta (1.5–3.5 Hz) Theta (3.5–7.5 Hz) Alpha 1 (7.5–9.5 Hz) Alpha 2 (9.5–12.5 Hz) Beta 1 (12.5–17.5 Hz) Beta 2 (17.5–30 Hz)	P4 is in phase, i.e., coincide, with theta, alpha 1, alpha 2 and beta 1, while delta and beta 2 go oppositely. E2 goes in phase with delta and theta and goes oppositely with alpha 2.

Table 4 (continued)

Study	N	Age	Hormonal assessment	Menstrual cycle phase	Harmonized menstrual phase terminology	Status of women	EEG recordings condition	EEG recording areas	Band frequency analyzed	Results
Bazanov et al. 2014	78	18–27	E2 and P4	Menstrual (from the first to the last day of menstruation) Follicular (from the next day after the end of menstruation to the beginning of increase in basal body temperature by 0.5–1 degree) Ovulatory (from the first day of increase in basal temperature to the first day of enhanced salivary progesterone activity) Luteal (from the first day of increase in salivary progesterone level to the beginning of its decrease) Premenstrual (from the first day of decrease in progesterone level to the beginning of menstruation)	M phase F phase O phase L phase Late L phase	Healthy women with regular menstrual cycles	At rest with eyes closed and eyes open	Frontal Central Parietal Occipital	Alpha 1 (7–10 Hz) Alpha 2 (10–13 Hz)	IAF and alpha band width are the maximum in the L phase. Salivary P4 levels are correlated positively with IAF and the power of the alpha 2 band, but negatively with the power of the alpha 1 band. The depth of power reduction in the alpha band in response to opening eyes is negatively related to P4 level.
Brötzner et al. 2014	57	20–28	E2 and P4	Early follicular (onset of menstruation plus five days) Late follicular (14 days before onset of menstruation) Luteal (three days post ovulation to five days before the onset of menstruation)	Early F phase Late F phase L phase	Healthy women with regular menstrual cycles	At rest with eyes closed	Parietal	Alpha (7–14 Hz)	Women in L phase have the highest and women in late F phase have the lowest IAF. IAF and center frequency is correlated negatively with E2 level, but IAF and center frequency is not correlated with P4.
Patnaik and Mohanty 2017	10	25–35	P4	Luteal (day 15–28 from the first day of bleeding) Follicular (day 1–14 from the first day of bleeding)	L phase Early and late F phases	Healthy women with regular menstrual cycles	At rest with eyes closed and eyes open	Frontal Central Occipital	Delta (0–4 Hz) Theta (4–8 Hz) Alpha (8–12 Hz) Beta (13–20 Hz)	Alpha absolute power at frontal, central, and occipital with eyes closed > eyes open condition during the early and late F phases. Delta absolute power with eye open at frontal, central, and occipital > eye closed condition during the L phase.

Table 4 (continued)

Study	N	Age	Hormonal assessment	Menstrual cycle phase	Harmonized menstrual phase terminology	Status of women	EEG recordings condition	EEG recording areas	Band frequency analyzed	Results
Summer et al., 2018	20	21–23	E2 and P4	Early follicular (1–5 of the menstrual cycle) Mid-luteal (20–25 of the menstrual cycle)	Early F phase L phase	Healthy women with regular menstrual cycles	At rest watching stationary gratings	Unspecified	Gamma (35–90 Hz)	The peak gamma frequency is higher in L phase vs. early F phase for stationary gratings stimuli. No significant correlation between P4 and peak gamma frequency for stationary gratings stimuli. No significant correlation between E2 and peak gamma frequency for stationary gratings stimuli.
Bazanov et al. 2017	59	17–29	P4 and cortisol	Menstrual (from the first to the last day of menstruation) Follicular (from the next day after the end of menstruation to the beginning of increase in basal body temperature by 0.5–1 degree) Ovulatory (from the first day of increase in basal temperature to the first day of enhanced salivary progesterone activity) Luteal (from the first day of increase in salivary progesterone level to the beginning of its decrease) Premenstrual (from the first day of decrease in progesterone level to the beginning of menstruation)	M phase Early and late F phases O phase L phase Late L phase	Healthy women with regular menstrual cycles	At rest with eyes closed and eyes open	Parietal	Alpha 1 (9–10.5 Hz) Alpha 2 (10.5–13 Hz)	Magnitude of alpha 1 amplitude suppression in response to open eyes is highest at early and late F phases. Duration of amplitude suppression in the alpha 1 range in response to open eyes is longest at L phase. Alpha 1 band width is wider at O phase than at M phase, early and late F phases, L phase, and late L phase. Alpha 2 band width is largest in L phase vs. M phase, early and late F phases, O phase, and late L phase. Alpha-bandwidth is increased during the L phase, which also corresponds to the highest level of salivary P4.

Table 4 (continued)

Study	<i>N</i>	Age	Hormonal assessment	Menstrual cycle phase	Harmonized menstrual phase terminology	Status of women	EEG recordings condition	EEG recording areas	Band frequency analyzed	Results
García-Granados et al. 2023	10	20–26	E2 and P4	Late F phase (days 11–16 after the first day of bleeding) Luteal phase (days 18–27 after of the first day of bleeding)	Late F phase L phase	Healthy women with regular menstrual cycles	At rest with eyes closed	Occipital	Alpha (8–13 Hz)	The IAF and absolute power of alpha in the late F phase < L phase.
Gaižauskaitė et al. 2024	68	18–35	E2, P4 and testosterone	Early follicular phase (3.2 day after menstruation) Mid- luteal phase (20.9 day after menstruation)	Early F phase L phase	Healthy women with regular menstrual cycles	At rest with eyes open	64-channel electrode cap positioned according to the international 10–20 system.	Alpha (8–13 Hz)	There is no significant difference in absolute power or in the IAF between early follicular and L phases.

E2 estradiol, *P4* progesterone, *M* menstrual, *F* follicular, *O* ovulatory, *L* luteal, *IAF* individual alpha frequency

Table 5 Quality ratings assigned to each study using the downs and black (DB) scale, and final rating according to menstrual cycle (MC) phase verification by hormonal assay test

Study	Reporting/6	External validity/1	Internal validity-bias/6	Internal validity- confounding (selection bias)/1	Power/2	Total score of DB/16 and a priori rating	Directness-MC phase verification by hormonal assay test?	Final rating
Dusser de Barenne and Gibbs, 1942	3	0	3	0	1	7; low	No	Verylow
Margerison et al. 1964	6	0	3	1	2	12; moderate	Yes	Moderate
Gautray, 1969	4	0	1	0	1	6; low	Yes	Low
Vogel et al. 1971	6	0	4	1	2	13; moderate	No	Low
Wuttke et al. 1975	5	1	4	0	2	12; moderate	Yes	Moderate
Creutzfeldt et al. 1976	5	1	5	1	2	14; high	Yes	High
Leary and Batho 1979	4	0	4	1	1	10; moderate	No	Low
Becker et al. 1982	6	1	5	1	2	15; high	Yes	High
Kaplan et al. 1990	6	0	4	1	2	13; moderate	No	Low
Solis-Ortiz et al., 1994	6	1	5	1	2	15; high	No	Moderate
Kaneda et al. 1997	6	0	4	0	2	12; moderate	No	Low
Krug et al. 1999	6	1	5	1	2	15; high	Yes	High
Solis-Ortiz et al. 2004	6	1	4	1	2	14; high	No	Moderate
Vasil'eva, 2005	5	0	3	0	2	10; moderate	No	Low
Solis-Ortiz et al., 2009	6	1	4	1	2	14; high	Yes	High
Bazanov et al. 2014	5	0	5	1	2	13; moderate	Yes	Moderate
Brötzner et al. 2014	6	1	5	1	2	15; high	Yes	High
Patnaik and Mohanty 2017	4	0	1	0	2	7; low	No	Verylow
Summer et al., 2018	6	1	5	1	2	15; high	Yes	High
Bazanov et al. 2017	6	1	5	1	2	15; high	Yes	High
Mulik et al. 2017	5	1	3	0	2	11; moderate	Yes	Moderate
García-Granados et al. 2023	6	1	5	1	2	15; high	Yes	High
Gaižauskaitė et al. 2024	6	1	5	1	2	15; high	Yes	High

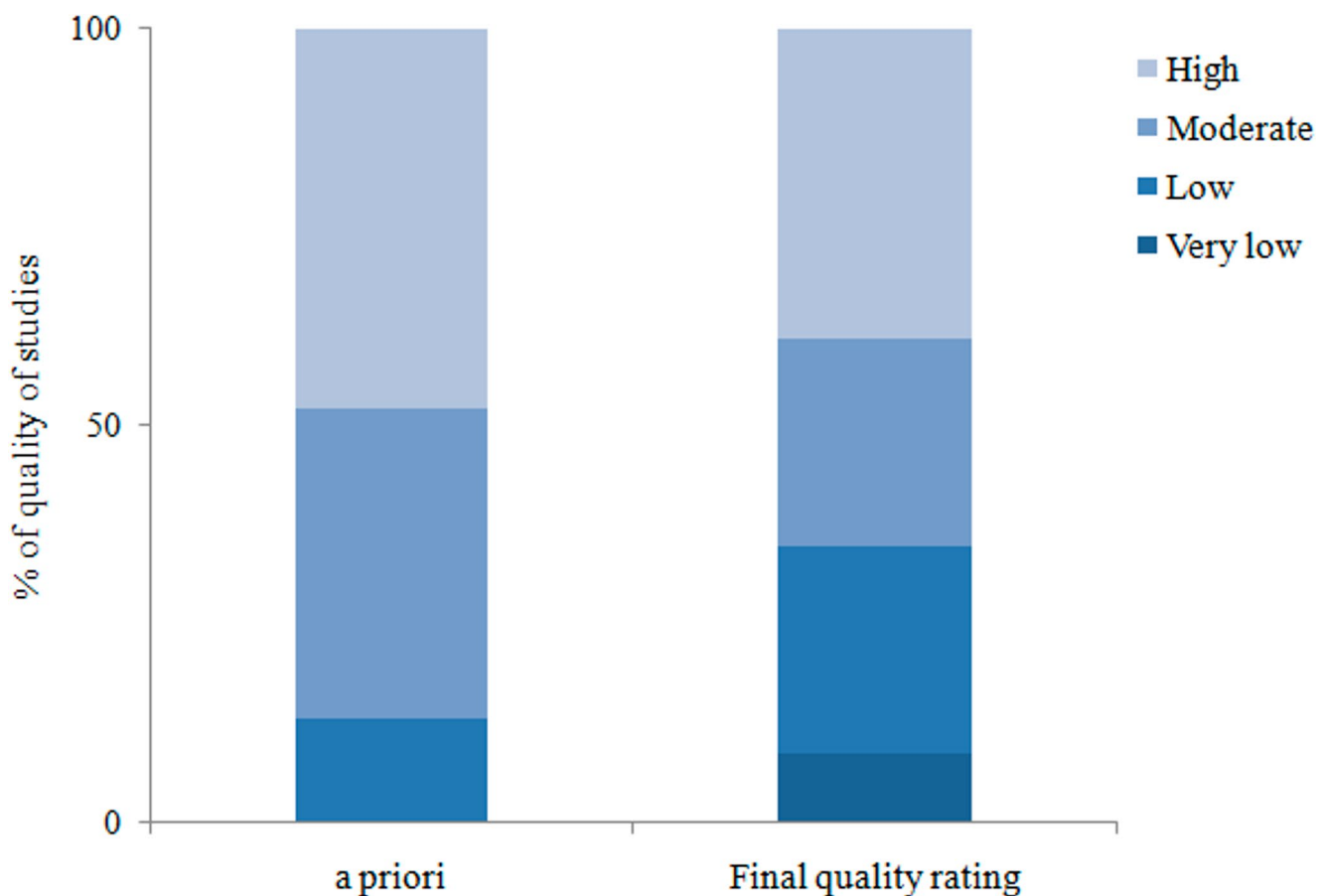


Fig. 3 Quality rating of outcomes from all included studies ($n=23$). Each bar represents the proportion of studies assigned a high, moderate, low, or very low quality rating. The x-axis represents the different

stages of the quality appraisal process, a priori and final quality rating (based on the question related to menstrual cycle phase verification by hormonal assay in each study)

Resting-state EEG activity during the menstrual phase

The IAF in women under eyes-closed condition was the lowest at the occipital area during the M phase (Mulik et al. 2017). Another study found that the relative power of delta at right parietal areas was higher during the M phase than during the O phase (Solis-Ortiz et al., 1994). Another work reported that the frequency of the theta band at the occipitocentral area was higher during the M phase than during the L phase (Becker et al. 1982). Meanwhile, the absolute power of alpha 1 and alpha 2 increased, and the absolute power of beta 1 decreased during the M phase when women were in the eyes-open condition (Solis-Ortiz et al. 2004).

Resting-state EEG activity during the follicular phase

A study showed that women under eyes-closed condition had a lower relative power of alpha in the posterior area during the F phase when compared with the O phase (Gautra,

1969). Another work found that the peak of alpha frequency at the occipital area in the F phase was lower than in the L phase (Kaplan et al. 1990). In addition, the absolute power of alpha 1 at the occipitocentral area drastically decreased during the F phase (Becker et al. 1982). Additionally, the relative power of the theta band in the left frontotemporal area was higher during the F phase than during the O phase (Gautray, 1969). Besides, the absolute power of alpha 0, beta 1, and beta 2 at occipitocentral areas was higher during the F phase than during the L phase (Becker et al. 1982). Other investigation also indicated that the maximum peak amplitude and the power of the alpha 1 and alpha 2 bands were higher during the F phase when compared to the L phase, late L phase, and M phase (Bazanov et al. 2014). However, Kaneda and colleagues (1997) did not find significant differences in the absolute power of any bands at the occipital area between the F phase and the L phase. The lowest IAF was observed during the late F phase (Brötzner et al. 2014). In addition, both IAF and absolute power of alpha were lower in the late F phase compared to the L phase (García-Granados et al. 2023).

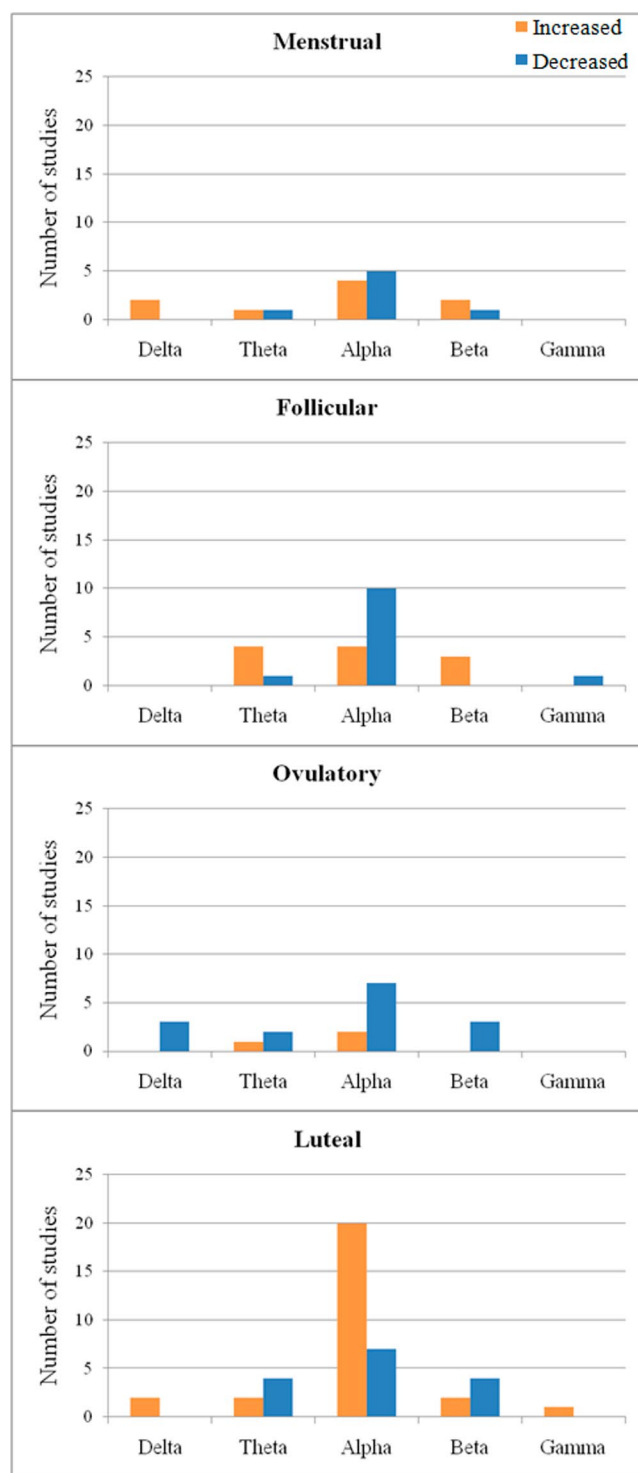


Fig. 4 Number of studies reporting increases or decreases in EEG activity across menstrual cycle phases

Regarding women under eyes-open condition, the alpha frequency at the bilateral occipital area was lower during the F phase than during the L phase (Vogel et al. 1971). Additionally, a study found that the highest amplitude suppression magnitude of the alpha 1 band occurred during the F

phase (Bazanova et al. 2017). Another study reported that the absolute power of alpha at frontal, central, and occipital areas was higher during the eyes-closed condition than during the eyes-open condition in the F phase (Patnaik and Mohanty 2017). However, one research reported no significant difference in the absolute power or in the IAF between early F phase and L phase (Gaižauskaitė et al. 2024).

Resting-state EEG activity during the ovulatory phase

Women under eyes-closed condition showed no significant differences in the alpha frequency at occipital and temporal areas between the O phase and late L phase (Margerison et al. 1964). However, a study found that the maximum peak amplitude and the power of the alpha 1 and alpha 2 bands were higher during the O phase than during the early and late L phase, and M phase (Bazanova et al. 2014).

EEG recordings under eyes-open condition revealed that the absolute powers of the delta and alpha 1 at the left occipital area, as well as the absolute power of beta 1 at the right central area, were lower during the O phase than during the late L phase. In addition, the absolute power of the alpha 1 band at the left frontal area, the beta 1 band at the right central and left parietal areas, the beta 2 band at the left central area, and the delta band at the occipital areas were lower during the O phase than during the M phase (Solis-Ortiz et al., 1994).

Resting-state EEG activity during the luteal phase

Women under eyes-closed condition showed the lowest IAF in the occipital area during the L phase when compared to the M phase (Mulik et al. 2017). On the other hand, other studies revealed that the alpha bandwidth (Bazanova et al. 2014) and the IAF (Brötzner et al. 2014; Bazanova et al. 2014) were the highest during the L phase. Another study found a similar result, because the weighted frequency of the alpha band in occipito-central and temporo-central areas was higher during the L phase than during the F phase (Creutzfeldt et al. 1976). Whilst the power and frequency beta band in the occipito-central and temporo-central areas did not significantly differ between the L phase and the F phase; however, the power of theta in the occipito-central and temporo-central areas was lower during the L phase than during the F phase (Creutzfeldt et al. 1976). Furthermore, the frequency of the theta band and the power of the alpha band in the temporo-central area were lower during the L phase than during the F phase (Creutzfeldt et al. 1976).

Concerning the eyes-open condition, the power of the alpha and theta in the frontal, central, and parietal areas during the L phase was lower than during the O phase and M

phase (Krug et al. 1999). Additionally, the absolute power of alpha 1 and alpha 2 decreased, while the absolute power of beta 1 increased during the early L phase. In an opposite way, the absolute power of alpha 1 and alpha 2 increased while the absolute power of beta 1 decreased during the late L phase (Solis-Ortiz et al. 2004). Other studies showed that the relative power of the alpha 1 band in the right central area and the alpha 2 band in the right frontal area was lower in the early L phase when compared with the late L phase (Solis-Ortiz et al., 1994). Another study revealed that the peak frequency of the gamma band was higher during the L phase than during the F phase (Summer et al., 2018). Lastly, the absolute power of the delta band in the frontal, central, and occipital areas was higher in the eyes-open condition in comparison with the eyes-closed condition during the L phase (Patnaik and Mohanty 2017).

Discussion

The present systematic review is the first that provides an integrated knowledge about the changes of spontaneous electrical activity of the brain at rest condition throughout the different phases of the healthy menstrual cycle. The major strength of the present review is based not only on resting-state EEG activity of the brain associated with the menstrual cycle phase but also includes a finer categorization of the findings based on estradiol and progesterone hormones (Table 4) and the quality ratings of the reviewed studies (Table 5; Fig. 3). Thus, these findings provide preliminary insights into the electrophysiological mechanisms that could underlie mood and cognitive variations across the menstrual cycle. The Fig. 4 shows the number of studies reporting increases or decreases in EEG activity across menstrual cycle phases. While some changes such as alpha and theta power differences between the F and L phase appear relatively consistent, results for delta, beta, and gamma bands remain inconclusive. Importantly, when incorporating the criterion of menstrual cycle phase verification via hormonal assay, the overall quality of evidence decreased. This highlights the critical role of accurate phase classification in ensuring methodological rigor and minimizing the risk of misclassification bias in menstrual cycle research. Many of the included studies are small, methodologically heterogeneous, and vary in EEG protocols, phase definitions, and hormonal measurement methods. Therefore, the interpretation of these findings must be approached with caution.

There are relatively few electrophysiological studies that have examined the changes of the EEG activity during each phase of the healthy menstrual cycle, and there are fewer studies available that have specifically investigated

the influence of sex hormones on electrical activity of the brain. In both situations, some results are inconsistent (e.g., Becker et al. 1982; Creutzfeldt et al. 1976; Solis-Ortiz et al., 1994). The inconsistency of EEG activity during the menstrual cycle could be due to methodological differences, such as the different determinations of the menstrual cycle phase and different approaches to EEG activity assessment. However, most of the EEG studies were focused on comparing the EEG activity between late F phase and L phase because these phases have the most contrasting levels of estradiol and progesterone between them. The late F phase is distinguished by the highest levels of estradiol and low progesterone, whereas the L phase is characterized by the highest level of progesterone and a lower secondary estradiol peak (Bull et al. 2019). While other studies considered additional phases. Another notable observation is that the majority of research focuses solely on how sex hormones affect alpha EEG activity. This is probably due to (1) the fact that alpha oscillation (7 to 13 Hz) is the most prominent EEG band in the resting state in healthy humans, particularly when the subject's eyes are closed (Berger 1935); (2) the methodological limitations of previous studies for analyzing other bands (i.e., delta, theta, beta, and gamma) without sophisticated techniques, as some of this research was conducted approximately 40–50 years ago; and (3) the hypothesis that alpha oscillation has a fundamental role in cognition (Jensen and Mazaheri 2010).

The EEG studies reviewed here tend to converge in showing reduced absolute power, frequency, and amplitude of alpha oscillations in frontal, parietal, and temporal regions during the M phase. These findings are corroborated by magnetoencephalography evidence indicating lower IAF and median frequency during the M phase compared to the F phase and L phase in the resting state (Haraguchi et al. 2021). Given the well-established association between alpha activity and cortical synchrony as well as functional connectivity—particularly within the default mode network (DMN)—this reduction in alpha power may serve as an electrophysiological marker of decreased within-network connectivity in the DMN. Such a disruption could contribute to an internal brain state that hampers cognitive performance during the M phase. The DMN is typically active during rest and has been implicated in self-referential thinking, introspection, and emotional-cognitive integration (Fransson, 2006). Efficient task performance often requires suppression of DMN activity to enable engagement of task-positive networks. In this context, neuroimaging studies have shown increased functional connectivity between the left frontal cortex and the DMN during the M phase, which has been associated with poorer task performance (Weis et al., 2017). Furthermore, depression and anxiety are known to correlate with reduced alpha power in all brain regions (Zhang et al.

2024), and it is well-documented that many women experience affective symptoms such as anxiety, depression, or sadness around menstruation. These emotional symptoms may be related to the observed reductions in alpha activity during the M phase. Nevertheless, while associations between alpha dynamics, DMN connectivity, and affective states have been proposed, the current evidence base is insufficient to establish causal or robust mechanistic pathways. These interpretations should therefore be viewed as hypothesis-generating rather than confirmatory.

Among the electrophysiological rhythms observed in the human brain, alpha oscillations (8–13 Hz) in occipital regions dominate during the eyes-closed resting states. This rhythm typically exhibits the highest amplitude under eyes-closed conditions and attenuates upon eye opening (a phenomenon known as alpha desynchronization), commonly interpreted as suppression of visual input or event-related desynchronization (Wan et al. 2019). Alpha activity is widely recognized as a marker of cortical processing and functional brain states (Bazanov et al. 2014). The studies reviewed here suggest that fluctuations in estradiol and progesterone levels across the menstrual cycle may influence resting-state EEG alpha activity. Specifically, during the early F phase, when both estradiol and progesterone are at their lowest, posterior alpha amplitude, absolute power, frequency, and IAF tend to appear reduced in some studies. As estradiol levels rise toward ovulation, a general trend toward increased frequency, amplitude, and absolute power of the alpha band, possibly reflects enhanced cortical synchrony. This period of elevated alpha activity, particularly in posterior regions, might correspond to a neurophysiological state characterized by optimal cognitive integration, spontaneous mental activity, and self-referential processing. During the L phase, when progesterone predominates and estradiol remains at moderate levels, alpha power and IAF continue to increase, possibly facilitating deeper levels of internal focus, introspection, and emotional processing. Notably, this heightened alpha activity—especially within the DMN—could underlie ruminative or self-referential thought patterns, which are known to be amplified in individuals predisposed to mood disturbances, such as anxiety or depression. In women with PMDD, this mechanism might contribute to the emotional instability, increased sensitivity, and dysphoric affect frequently reported during the L phase. Nonetheless, these interpretations should be viewed cautiously given methodological variability and the limited number of studies directly linking hormones, alpha activity, and mood.

At the same time, some studies reviewed here reported that absolute alpha power in the frontal and parietal regions is lower during the F phase compared to the L phase. These brain regions are critically involved in top-down attentional control and cognitive regulation (Behrmann et al. 2004). It is

suggested that alpha oscillations may support top-down regulation and higher-order information integration by providing temporal frameworks for neuronal connections via the timing of functional inhibition (Klimesch 2012). According to Moretti et al. (2004), alpha activity during wakeful rest reflects the functional integrity of the cholinergic system and cortico-cortical connectivity. Therefore, reduced alpha power in these areas during the F phase—particularly in the late F phase when estradiol peaks—may indicate increased desynchronization and enhanced engagement of attentional networks. This neurophysiological state, driven by elevated estradiol, may contribute not only to enhanced cognitive performance but also to improved affective regulation, as frontal and parietal alpha asymmetry has been associated with positive mood and emotional stability (Liparoti et al. 2021). Avila-Varela et al. (2024) demonstrated that cognitive performance, particularly in information processing tasks, peaks near ovulation and declines progressively throughout the L phase. Furthermore, Pritschet et al. (2020) reported that estradiol enhances intra-network connectivity within both the DMN and dorsal attention network, especially during the late F phase, reinforcing the notion that cyclical hormonal dynamics modulate both large-scale brain networks and affective-cognitive integration across the menstrual cycle. In contrast, the elevated alpha power observed during the L phase may indicate a disengagement from externally oriented attentional processes and a shift toward internally directed cognition. This shift, in the context of rising progesterone and relatively lower estradiol levels, may underlie increased emotional sensitivity, vulnerability to negative affect, and diminished cognitive flexibility. Indeed, frontal alpha activity has been linked to affective states, with higher alpha power reflecting reduced cortical activation and greater risk for depressive or anxious mood (Xie et al. 2023). This may partially explain the heightened prevalence of mood lability, rumination, and emotional dysregulation reported during the L phase, especially among women with premenstrual mood symptoms.

With respect to the IAF, some studies examined here report that the IAF tends to be higher in the L phase and lower in the late F phase, showing a negative correlation with estradiol and a positive correlation with progesterone. Although higher IAF is generally associated with enhanced cognitive processing and attentional efficiency, its increase during the L phase—when many women report cognitive and emotional challenges—may reflect a more complex interaction between cortical rhythms and hormonal modulation. It is possible that progesterone, through its GABAergic effects, modulates neural excitability in way that contribute to reduced cognitive flexibility or affective dysregulation, even in the presence of elevated IAF. Conversely, the lower IAF observed during the late F phase coincides with

peak cognitive and emotional functioning, possibly due to estradiol-driven enhancement of network integration. These findings remain tentative and further research is needed to clarify whether IAF consistently reflects sensitivity to hormonal changes or potential vulnerability to mood disturbances across the menstrual cycle.

Finally, some studies indicate that theta frequency and theta absolute power in temporo-central and occipito-central areas, as well as beta absolute power, tend to be higher during the F phase compared to the L phase. In contrast, gamma peak frequency appears to be lower in the F phase compared to the L phase. These findings suggest that the late F phase may be associated with enhanced neural states favorable for memory-related processes. Theta oscillations, particularly those generated in hippocampal circuits, are critically involved in memory encoding and retrieval (Fuentemilla et al. 2014). Moreover, theta activity is tightly coupled with the DMN, which supports episodic memory (Jeong et al. 2015) and autobiographical recall (Lin et al. 2016). Thus, an increase in theta activity during the late F phase, possibly associated with elevated estradiol, may contribute to a more optimal neurophysiological state for episodic and autobiographical memory function. This phase-specific enhancement may underlie the improved performance on verbal and associative memory tasks observed during periods of high estradiol (Hausmann et al. 2000). Similarly, increased beta power, typically associated with motor planning and cognitive control (Engel and Fries 2010), may reflect enhanced top-down modulation during the late F phase. The observed decrease in gamma peak frequency during the same phase—an oscillatory band implicated in sensory integration, motor preparation, and conscious awareness (Leopold et al., 2003)—might indicate a phase-specific shift in cortical excitability or processing demands. The gamma band is associated with the integration of the stimuli, movement preparation, and consciousness (Leopold et al., 2003). All of these electrophysiological results suggest that the late F phase, characterized by rising estradiol levels, could promote a more integrated neurophysiological state associated with cognitive performance. This interpretation is tentatively supported by behavioral evidence indicating that rising estradiol concentrations are positively associated with improvements in verbal memory tasks, working memory, and paired-associate learning but negatively with performance in spatial tasks (Hausmann et al. 2000; Rosenberg and Park 2002).

Conclusion

This systematic review suggests that resting-state EEG activity varies across the menstrual cycle, with some evidence indicating that these fluctuations are influenced by

hormonal changes. The most consistent changes appear in the theta and alpha bands, particularly during phases of pronounced hormonal variation, i.e., around ovulation and the L phase. These oscillatory dynamics could be linked to changes in emotional reactivity and affective symptoms, potentially contributing to the neurobiological basis of conditions like PMDD. The present findings underscore the importance of considering menstrual cycle phase and hormonal status in studies involving women's brain function and mental health. Nonetheless, the conclusions presented here should be viewed as preliminary models that generate testable hypotheses, rather than as definitive or generalizable patterns. Further research is needed to establish the reliability of these electrophysiological trends.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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